

MICHIGAN DEPARTMENT OF ENVIRONMENTAL QUALITY

INTEROFFICE COMMUNICATION

TO: Daniel Dell, Lakes Michigan and Superior Permits Unit, Permits Section
Water Bureau

FROM: Doyle Brunsen, Surface Water Assessment Section, Water Bureau

DATE: July 31, 2007

SUBJECT: Wexford County Landfill
National Pollutant Discharge Elimination System (NPDES) Permit
Effluent Limits Only (ELOs)

We have completed our evaluation for ELOs for the planning purposes for an NPDES permit for the Wexford County Landfill. They propose to continuously discharge 0.1 million gallons per day (MGD) (0.15 cubic feet per second [cfs]) of treated landfill leachate to an unnamed tributary to Manton Creek (NE 1/4, NE 1/4, Section 34, R9W, T23N) in Wexford County.

Manton Creek is protected for coldwater fish and other indigenous aquatic life and wildlife, fish consumption, partial body contact recreation, total body contact recreation from May through October, agriculture, navigation, industrial water supply, and public water supply at the point of intake.

The monthly 50 percent (%) and 95% exceedance flows, in cfs, for Manton Creek at this location are as follows:

	<u>JAN.</u>	<u>FEB.</u>	<u>MAR.</u>	<u>APR.</u>	<u>MAY</u>	<u>JUNE</u>
95%	3.7	3.4	5.3	9.7	4.2	2.8
50%	7.2	7.4	15	25	13	6.2
	<u>JULY</u>	<u>AUG.</u>	<u>SEP.</u>	<u>OCT.</u>	<u>NOV.</u>	<u>DEC.</u>
95%	2.5	2.1	2.7	4.7	4.7	4.6
50%	4.5	3.9	4.6	7.0	10	8.3

The unnamed tributary to Manton Creek is ephemeral with a 95% and 50% exceedance flow of 0 cfs flow all year.

RECOMMENDATIONS

The recommended limits are found in Table 1. For the period from October through April, we recommend a 5-day Carbonaceous Biochemical Oxygen Demand (CBOD₅) of 60 milligrams per liter (mg/l) as a daily maximum, and dissolved oxygen (D.O.) at 5 mg/l as a daily minimum are sufficient to protect the receiving stream. An ammonia limit of 30 mg/l as a daily maximum is recommended for the December – March period. These recommendations meet the D.O. water quality standard (WQS) of 7 mg/l as a daily minimum and are preliminary and may change based on the NPDES permit application submitted by the Wexford County Landfill. Ammonia toxicity recommendations are based on the 0.029 mg/l un-ionized ammonia toxicity criterion at the requested discharge flow.

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We evaluated the effects of the discharge on the unnamed tributary to Manton Creek using a Streeter-Phelps D.O. model. A diurnal variation of 0.65 mg/l was used as the unnamed tributary.

The recommendations contained herein are based on WQS. Treatment methods or economic concerns have not been addressed. We do not imply in these recommendations that other methods or considerations should not be used in deciding the permit limits.

db:rm

Attachment

cc/att: Mike Stifler, Cadillac District Supervisor, Water Bureau (WB)
Alvin Lam, Permit Section, WB
Brenda Sayles/Effluent Limit File, WB

TABLE 1

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FACILITY: Wexford County Landfill
DISCHARGE: Continuous
FACILITY DESIGN FLOW: 0.1 MGD (0.15 cfs)
RECEIVING STREAM: Manton Creek

PARAMETER	TIME PERIOD	EFFLUENT		COMMENT	RATIONALE
		CONC mg/l	MASS lb/d		
D.O.	ALL YEAR	5.0		DAILY MINIMUM	ACUTE D.O.
CBOD ₅	MAY 1-SEP 30	10	8.3	DAILY MAXIMUM	D.O. STANDARD
	OCT 1-APR 30	60	50	DAILY MAXIMUM	D.O. STANDARD
NH ₃ -N	MAY 1-SEP 30	3	2.5	DAILY MAXIMUM	D.O. STANDARD
	MAY 1-SEP 30	2.2	1.8	30-DAY AVERAGE	CHRONIC TOXICITY
	OCT 1-NOV 30	14	11.7	DAILY MAXIMUM	D.O. STANDARD
	OCT 1-NOV 30	19	15.9	30-DAY AVERAGE	CHRONIC TOXICITY
	DEC 1-MAR 31	7.4	6.2	30-DAY AVERAGE	CHRONIC TOXICITY
	APR 1-APR 30	20	16.7	DAILY MAXIMUM	D.O. STANDARD
	APR 1-APR 30	4.6	3.8	30-DAY AVERAGE	CHRONIC TOXICITY